Basic Linear Regression: Exercises

Recall

Dataset $D = \{(x^{(i)}, y^{(i)})\}_{i=1}^{N}$, model $f(x \mid m, c) = mx + c$, and loss

$$L_2(m, c) = \frac{1}{N} \sum_{i=1}^{N} \Big(f(x^{(i)} \mid m, c) - y^{(i)} \Big)^2.$$

L_2 is smallest at

$$m = \frac{\overline{xy} - \overline{x} \cdot \overline{y}}{\overline{x^2} - \overline{x}^2} \qquad c = \overline{y} - m\overline{x}$$

where

$$\overline{x} = \frac{1}{N} \sum_{i=1}^{N} x^{(i)} \qquad \overline{y} = \frac{1}{N} \sum_{i=1}^{N} y^{(i)}$$

$$\overline{xy} = \frac{1}{N} \sum_{i=1}^{N} x^{(i)} y^{(i)} \qquad \overline{x^2} = \frac{1}{N} \sum_{i=1}^{N} \Big(x^{(i)} \Big)^2$$

Exercise 1: Two Points

Let $D = \{(x^{(1)}, y^{(1)}), (x^{(2)}, y^{(2)})\}$ with $x^{(1)} \neq x^{(2)}$. Using the formulas above, show that

$$m = \frac{y^{(2)} - y^{(1)}}{x^{(2)} - x^{(1)}} \qquad c = \frac{x^{(2)} y^{(1)} - x^{(1)} y^{(2)}}{x^{(2)} - x^{(1)}}$$

What does this tell you about the line of best fit for just two points?

Hint: what is $L_2(m, c)$?

Exercise 2: $\frac{\partial L_2}{\partial c}$

Show that

$$\frac{\partial L_2}{\partial c} = \frac{2}{N} \sum_{i=1}^{N} \Big(f(x^{(i)} \mid m, c) - y^{(i)} \Big)$$

Hint: follow the same steps as $\frac{\partial L_2}{\partial m}$ in the slides.

Exercise 3: Deriving m and c

Starting from

$$0 = \frac{2}{N} \sum_{i=1}^{N} \Big(f(x^{(i)} \mid m, c) - y^{(i)} \Big) x^{(i)}$$

$$0 = \frac{2}{N} \sum_{i=1}^{N} \Big(f(x^{(i)} \mid m, c) - y^{(i)} \Big)$$

derive the formulas for m and c in the Recall section.

Hint: write each sum as a mean, e.g. $\frac{1}{N} \sum_{i=1}^{N} x^{(i)} y^{(i)} = \overline{xy}$. Then solve the second equation for c and substitute it into the first.